

Question Paper

Exam Date & Time: 25-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Embedded programming [ECE_4450]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Discuss and illustrate the role of a program in an Embedded Systems. (4)
A) Course Outcome: "CO1" / Cognitive Level: "Understand"
B) Extend the role of Compiling, Linking and Locating in an Embedded Systems. (3)
Course Outcome: "CO2" / Cognitive Level: "Application"
C) Elaborate the significance of a Simulator in an Embedded Program. (3)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) With an example, explain the salient features of Serial ports. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Application"
B) Discuss the characteristics of Flash memory used. (3)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
C) Discuss the techniques used for reducing a memory size. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 3) Elaborate on the impact of OOP in an Embedded OS. (4)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) List and explain different signals used in DMA Controller. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
C) List and explain the built-in functions used in Microprocessors. (3)
Course Outcome: "CO3" / Cognitive Level: "Application"
- 4) With a pseudocode, discuss the salient features of Round robin architecture without interrupts. (4)
A) Course Outcome: "CO4" / Cognitive Level: "Application"

B) With a neat diagram, extend the significance of Priority in RTOS architecture. (3)

Course Outcome: "CO4" / Cognitive Level: "Analysis"

C) With a suitable pseudocode, discuss how Semaphores protects data. (3)

Course Outcome: "CO4" / Cognitive Level: "Application"

5) With a neat diagram, explain the process of building application software using native tools. (4)

A) Course Outcome: "CO5" / Cognitive Level: "Application"

B) With a typical test system, explain the goals of testing process defined in Embedded Systems. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

C) Differentiate between the poor plan and better plan for testing. (3)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4450:Embedded programming

#	Course Outcome	Description	Marks
1	1	CO 1	4
2	2	CO 2	6
3	3	CO 3	20
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 ECE_4450:Embedded programming

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	3	10(20%)
3	Application	8	27(54%)
4	Analysis	4	13(26%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		15	50 (100%)